Reflection with Action-Induced Visual Differences for Desktop GUI Agents

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Paper version: arXiv 2608.24015v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Existing reflectors collapse change detection and outcome verification into one step, leaving evidence implicit and yielding weakly grounded decisions.*

Desktop screens change subtly or all at once; the one-pass POR reflector misses or misreads both — 61 of 223 failed GUI-Owl-32B trajectories.

Why one-pass reflection breaks

1. **27.35%** of failures: reflector (GUI-Owl-32B)
2. **70.49%** of those: wrong diff identification
3. one pass couples “what changed” + “did it work” — evidence stays implicit

Evaluation setup

OSWorld-Verified · 361 | WindowsAgentArena · 154 | 4 backbones | 3 runs, mean | steps ≤ 50 / ≤ 15 | online only

- **Both online**; no offline metric reported.
- POR per Mobile-Agent-v3; reflector swapped.

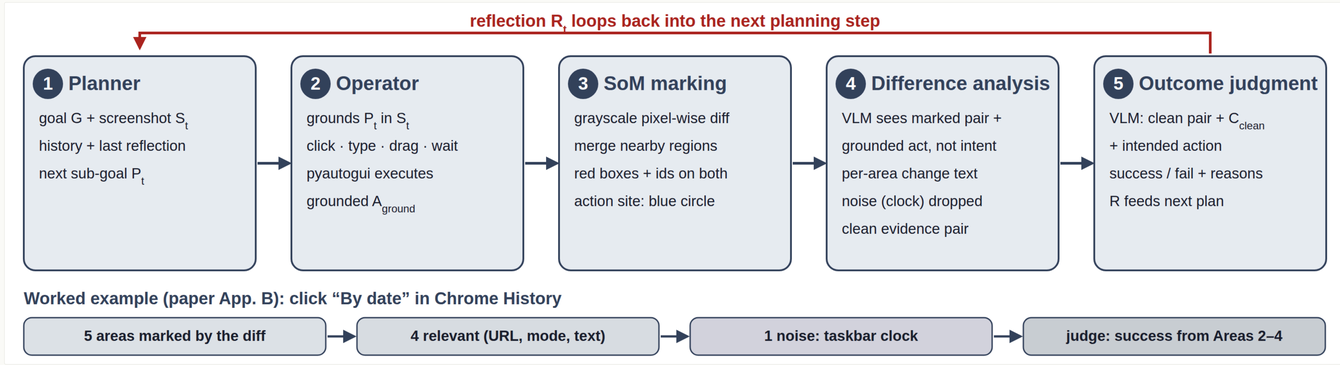
Our read (CUA Radar)

61	★★	online
score of 100	band	evidence level

- **Reports:** online eval; 3-run variance; matched budgets; ablation; reflector audit; cost. **Not reported:** held-out env; code.
- **Flags:** AP4 — judge used without two-sided reliability validation.
- Pattern C18 (post-action reflection). Authors’ numbers, not rerun.

Method: Evidence-First Reflection (EFR), a two-stage reflector

- **Sees / acts:** raw pre- and post-action screenshots; pyautogui actions (click, type, drag, hotkey).
- **New:** CV pixel-diff + SoM marks changed regions; VLM describes and filters them as explicit evidence, then judges.
- **Signal:** none learned — a prompt-level reflector swap inside the POR loop.

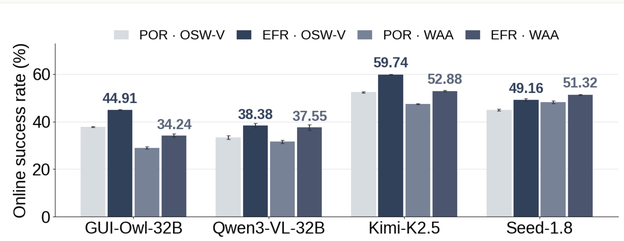


Our redraw (not the paper’s figure): the POR loop with EFR as reflector, and the paper’s worked evidence-filtering example.

$$R = f_{\text{judge}}\left(S_{\text{pre}}^{\text{clean}},\ S_{\text{post}}^{\text{clean}},\ C^{\text{clean}},\ A_{\text{intend}}\right)$$

Stage 2: the verdict comes from cleaned evidence plus the intended action (Eq. 7).

Key results: online success rate (%)



Tables 1–2, redrawn: 3-run means, overall success; std bars as reported.

+5.94 avg on OSW-V, +4.95 on WAA, all 4 backbones

SO WHAT → *OSW-V: 361 tasks; WAA: 154 tasks. Both online.*

Ablation: OSW-V success rate (%)

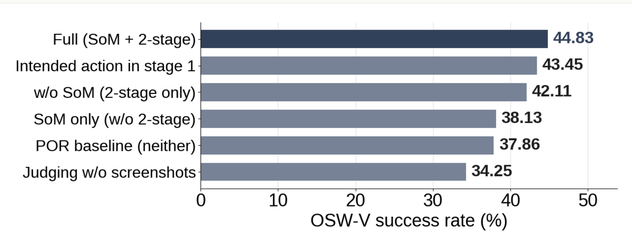


Table 3, redrawn: GUI-Owl-32B on OSWorld-Verified.

SO WHAT → *Judging without screenshots −10.58 and skipping diff identification −6.70 hurt most. Grounded (not intended) action in stage 1: −1.38. SoM alone: −2.72. POR baseline itself: −6.97. Every variant sits below the full method.*

Headline Numbers

+5.94
avg gain on OSW-V
+4.95 WAA

59.74% best (Kimi-K2.5)
+7.11 reflector accuracy
0.86 Pearson r

The paper’s own controls

CONTROL	RESULT	CONTEXT
Steps ≤ 50 / ≤ 15	same for POR, EFR	spec. vs OSS caps
Reflector swapped	POR (Mobile-Agent-v3)	4 backbones
Manual audit	acc. 80.39→87.50	Pearson r = 0.86
Per-task cost	+15.8% time +14.9% tok	+13.3% / +17.9% WAA
Flips (OSW-V)	12.68% fail→success	3.27% success→fail

Takeaway

SoM diffs + two-stage judging: +5.94 avg on OSW-V, +4.95 on WAA, for +15.8% latency.

reflect. acc. +7.11 | fail-attr. 27.35→13.57% | steps −11.5% on OSW-V | r = 0.86

Scan to read the full paper

arxiv.org/abs/2608.24015v1